

INVOLVE: An AI-Powered On-Demand Local Services Platform with Intelligent Issue Diagnosis and Emergency-Based Worker Assignment

ABSTRACT

In today's fast-paced world, people frequently encounter unexpected situations such as vehicle breakdowns, electrical faults, plumbing failures, urgent grooming requirements, and home maintenance emergencies that require immediate professional assistance. However, locating a reliable and available service provider within a short time remains a significant challenge due to delayed responses, limited service availability, lack of pricing transparency, and poor communication. Existing service platforms primarily focus on scheduled bookings and often fail to address emergency situations effectively. They also lack intelligent problem assessment, real-time tracking, and efficient worker allocation, resulting in longer waiting times and reduced customer satisfaction.

INVOLVE is an AI-powered on-demand local services platform designed to bridge this gap by providing fast, reliable, and intelligent service delivery. The platform integrates AI-based issue diagnosis using user descriptions and images to identify problems accurately and estimate their severity. Based on the emergency level, worker expertise, availability, proximity, ratings, and historical performance, the system intelligently assigns the most suitable verified professional instead of simply selecting the nearest one. Additional features such as real-time GPS tracking, ETA prediction, dynamic pricing, and blockchain-secured payments further enhance transparency, security, and user trust. By ensuring a guaranteed 10-minute response time for emergency requests, INVOLVE aims to improve service efficiency, minimize response delays, and deliver a seamless, technology-driven experience for both customers and service professionals.

Keywords:

- Artificial Intelligence (AI)
- On-Demand Local Services
- Emergency Worker Assignment
- Real-Time GPS Tracking
- Blockchain-Based Payments

INTRODUCTION

The rapid advancement of digital technology has transformed the way people access everyday services, making them faster, more convenient, and easily accessible. Despite this progress, obtaining immediate assistance during emergency situations such as vehicle breakdowns, electrical failures, plumbing leaks, appliance malfunctions, urgent home maintenance, and personal care emergencies remains a significant challenge. In such situations, users require quick access to reliable professionals, but existing service platforms are primarily designed for scheduled appointments rather than urgent requests. As a result, users often experience delayed responses, uncertainty regarding service quality, lack of real-time communication, and limited visibility into the progress of their service requests.

The emergence of Artificial Intelligence (AI), cloud computing, real-time communication technologies, and GPS-based location services provides an opportunity to improve the efficiency of emergency service delivery. These technologies enable intelligent decision-making by analysing user requests, identifying service requirements, estimating the severity of issues, and selecting the most appropriate service professional. Instead of assigning workers solely based on proximity or availability, AI-powered systems can evaluate multiple factors such as worker expertise, historical performance, availability, workload, customer ratings, and estimated arrival time. This intelligent approach improves service accuracy, minimizes waiting time, and enhances customer satisfaction.

Although several online platforms connect users with local service providers, they still suffer from important limitations. Most require advance bookings or depend on professionals to manually accept service requests, making them unsuitable for emergency situations. Many platforms lack intelligent problem diagnosis, transparent pricing, verified worker information, and real-time tracking, leading to inefficient worker allocation and inconsistent service quality. Users often find it difficult to explain technical issues accurately, while service professionals receive insufficient information before reaching the location. Consequently, professionals may arrive without the necessary expertise or equipment, increasing service delays and operational costs. Factors such as traffic congestion, uneven distribution of professionals, and unpredictable demand further reduce the effectiveness of traditional service allocation methods.

To overcome these challenges, this project proposes INVOLVE – An AI-Powered On-Demand Local Services Platform, an intelligent solution designed to provide fast, reliable, and technology-driven local services. The platform uses AI-based issue diagnosis to analyse user descriptions and uploaded images, enabling accurate identification of service requirements and emergency severity. Based on this analysis, the system intelligently ranks and assigns verified professionals by considering expertise, availability, experience, proximity, workload, customer ratings, and emergency priority. Rather than simply selecting the nearest worker, the platform ensures that the most suitable professional is assigned to each request, improving both response quality and efficiency.

The proposed platform also incorporates advanced features such as real-time GPS tracking, intelligent Estimated Time of Arrival (ETA) prediction, emergency-priority handling, dynamic

pricing, secure blockchain-based payment processing, and continuous service status updates. These features improve transparency, strengthen user trust, and provide a seamless experience for both customers and service professionals. By integrating Artificial Intelligence with modern communication and location technologies, INVOLVE aims to reduce response time, optimize worker allocation, improve service reliability, and enhance overall customer satisfaction. The platform represents a significant step toward building a smarter, faster, and more efficient ecosystem for on-demand local emergency services.

2. LITERATURE REVIEW

The rapid advancement of digital technology has significantly transformed the local service industry by enabling users to book professionals through online platforms. Applications providing home maintenance, repair, beauty, vehicle assistance, and appliance servicing have replaced traditional methods of searching for local workers through personal contacts or offline agencies. Platforms such as Urban Company, TaskRabbit, and Thumbtack have demonstrated that digital marketplaces can improve accessibility, transparency, and convenience by connecting customers with verified service providers. These platforms offer features such as online booking, customer reviews, digital payments, and service tracking. Although they have simplified the process of obtaining local services, they primarily focus on scheduled appointments and are not specifically designed to handle emergency situations requiring immediate assistance.

Several researchers have emphasized the importance of Artificial Intelligence (AI) in enhancing service management systems. AI techniques such as machine learning, recommendation systems, and predictive analytics enable intelligent analysis of user requests and support automated decision-making. Existing studies show that AI can evaluate multiple parameters, including worker expertise, availability, customer ratings, historical performance, and workload, to recommend suitable professionals. Compared with traditional first-come-first-served or nearest-worker allocation methods, AI-based worker selection improves assignment accuracy and reduces service delays. However, many commercial platforms continue to rely mainly on location-based matching, limiting their ability to provide the most qualified professional for complex or urgent service requests.

Recent developments in Natural Language Processing (NLP) have further expanded the capabilities of intelligent service platforms. NLP techniques allow systems to analyse user descriptions, identify the nature of reported problems, and classify service requests automatically. Instead of relying entirely on manual categorization, AI-based text analysis improves the accuracy of service identification while reducing communication gaps between customers and professionals. Researchers have demonstrated that NLP-based issue classification significantly improves response efficiency, particularly when users lack technical knowledge to explain problems accurately.

Computer vision has also emerged as an important technology for service diagnosis. Deep learning models trained on images can identify visible faults such as plumbing leaks, electrical damage, vehicle defects, cracked appliances, or structural problems. Image-based diagnosis provides additional information that helps professionals estimate service

complexity before reaching the customer's location. Research suggests that combining image analysis with textual descriptions improves diagnostic accuracy and reduces unnecessary repeat visits. Despite these advantages, most existing local service platforms still depend mainly on manual descriptions without integrating intelligent image analysis into their booking systems.

Location-aware technologies have become essential components of modern on-demand service applications. GPS-based tracking enables platforms to identify nearby professionals, monitor worker movement, estimate arrival time, and provide customers with real-time service updates. Several studies have shown that integrating traffic information, road conditions, and predictive Estimated Time of Arrival (ETA) algorithms improves route optimization and minimizes customer waiting time. Nevertheless, many existing systems assign workers solely based on distance while ignoring expertise, workload distribution, emergency severity, and historical service quality, resulting in inefficient resource utilization.

Dynamic pricing has also received considerable attention in recent research. Pricing models based on demand, service complexity, urgency, travel distance, and resource availability help maintain fairness for both customers and service providers. Intelligent pricing algorithms ensure transparent cost estimation before service confirmation, improving customer trust and reducing pricing disputes. However, many current service platforms provide fixed or manually estimated prices that fail to reflect real-time operating conditions and emergency priorities.

Another important area of research involves secure digital payment systems. Blockchain technology has emerged as a reliable solution for protecting financial transactions through decentralized and tamper-resistant record management. Researchers have highlighted the advantages of blockchain in ensuring payment transparency, preventing fraud, maintaining transaction integrity, and increasing trust among stakeholders. Although blockchain has been widely studied in banking, healthcare, and supply chain management, its implementation in local service platforms remains relatively limited, particularly for emergency service applications.

The proposed project, INVOLVE – An AI-Powered On-Demand Local Services Platform, builds upon these research findings by integrating multiple advanced technologies into one unified solution. The platform combines AI-based problem diagnosis, image-assisted issue identification, intelligent worker ranking based on expertise, availability, workload, and emergency priority, real-time GPS tracking, ETA prediction, dynamic pricing, and blockchain-enabled secure payments. Unlike conventional platforms that primarily emphasize scheduled bookings, INVOLVE focuses on rapid emergency response and intelligent decision-making. By selecting the most suitable verified professional rather than merely the nearest available worker, the platform aims to reduce response time, improve service reliability, optimize resource utilization, and enhance overall customer satisfaction. This integrated approach addresses the shortcomings identified in previous studies and contributes toward the development of a smarter, faster, and more dependable local emergency service platform.

3. Problem Statement & Objectives

Problem Statement

In today's fast-paced lifestyle, people frequently encounter emergency situations such as vehicle breakdowns, electrical faults, plumbing leaks, appliance failures, and urgent home maintenance that require immediate professional assistance. However, existing local service platforms primarily focus on scheduled bookings and often fail to provide rapid responses during emergencies. Users experience long waiting times, difficulty finding qualified professionals, lack of transparent pricing, limited real-time tracking, and inefficient communication. Moreover, current systems generally assign workers based only on proximity or availability without considering factors such as expertise, workload, emergency severity, and historical performance. The absence of AI-based issue diagnosis further increases the chances of incorrect service allocation, repeated visits, and delayed problem resolution. These limitations reduce service efficiency, customer satisfaction, and overall reliability. Therefore, there is a need for an intelligent, technology-driven platform that can accurately identify user problems, prioritize emergency requests, and assign the most suitable verified professional while ensuring transparency, security, and real-time communication throughout the service process.

Objectives of the Study

The primary objective of this project is to design and develop **INVOLVE – An AI-Powered On-Demand Local Services Platform** that delivers fast, reliable, and intelligent emergency assistance. The specific objectives are:

- To provide instant service booking with minimal response time.
- To implement AI-based issue diagnosis using user descriptions and images.
- To intelligently assign verified professionals based on expertise, availability, emergency priority, and location.
- To enable real-time GPS tracking and ETA prediction for service monitoring.
- To ensure transparent dynamic pricing and secure blockchain-based digital payments.
- To offer a user-friendly, reliable, and 24×7 emergency service platform for customers and professionals.

4. Proposed Methodology

The proposed methodology presents the design and implementation approach of INVOLVE – An AI-Powered On-Demand Local Services Platform, developed to provide fast, reliable, and intelligent emergency assistance for local service requests. The methodology integrates Artificial Intelligence (AI), cloud computing, real-time communication, geolocation services, and blockchain technology to create a unified platform that improves response time, service quality, and operational efficiency. Unlike conventional service applications that simply

connect users with nearby professionals, the proposed system analyses the nature of the reported problem, determines its urgency, and intelligently assigns the most suitable verified worker based on multiple evaluation parameters.

The service process begins when a user submits a request through the web or mobile application. Users can describe the issue using text, upload one or more images, or provide voice input for better explanation. The platform collects the user's location through GPS and securely transmits the request to the server. This information serves as the input for the AI engine, which performs preliminary issue analysis before assigning a professional.

The Artificial Intelligence module plays a central role in the proposed methodology. It analyses the user's description and uploaded images to identify the probable problem, estimate its severity, and classify the request into the appropriate service category, such as plumbing, electrical repair, appliance maintenance, vehicle assistance, or home services. The AI module also determines whether the request should be treated as an emergency or a normal service request. This intelligent diagnosis reduces communication errors, helps users describe technical problems more accurately, and enables professionals to prepare the necessary tools and equipment before arriving at the service location.

Once the service category and priority level are determined, the worker assignment module searches for all verified professionals available within the service area. Instead of assigning only the nearest worker, the system evaluates multiple parameters, including professional expertise, certification, service experience, customer ratings, historical performance, current workload, availability, travel distance, estimated arrival time, and emergency priority. Each professional is assigned a suitability score based on these factors. The worker with the highest score is selected to ensure that users receive assistance from the most qualified professional rather than simply the closest one. This intelligent allocation improves service quality, reduces repeated visits, and optimizes workforce utilization.

After successful assignment, both the customer and the service professional receive instant notifications through the real-time communication system. Technologies such as Socket.IO and Redis enable immediate synchronization of assignment status, service progress, and location updates between all participants. Customers can monitor the professional's live location using GPS tracking, while the system continuously calculates the Estimated Time of Arrival (ETA) by considering current traffic conditions and route optimization. Real-time updates improve transparency, reduce uncertainty, and allow users to prepare before the professional arrives.

The proposed methodology also incorporates a dynamic pricing mechanism that estimates service charges based on service category, complexity of the issue, travel distance, emergency priority, and expected repair effort. Users receive a transparent cost estimate before confirming the booking, minimizing pricing disputes and improving trust. Following successful service completion, payments are processed through a blockchain-enabled payment system that securely records transaction details. Blockchain technology ensures transparency, prevents transaction manipulation, and provides secure digital payment verification for both customers and service professionals.

Administrators play an important role in maintaining the platform. They verify worker registrations, approve certifications, monitor service quality, resolve customer complaints, manage service categories, and analyse system performance using service analytics. Historical service records are stored securely within the database and can be used by the AI module to continuously improve diagnosis accuracy, worker recommendation, and emergency response strategies through machine learning techniques.

The proposed methodology follows a modular architecture consisting of the presentation layer, application layer, AI layer, real-time communication layer, and security layer. This layered architecture improves scalability, maintainability, and future expansion while ensuring secure communication through REST APIs and encrypted data transfer. The system is designed to support increasing numbers of users and professionals without affecting performance.

Overall, the proposed methodology provides an intelligent framework for modern emergency local services by combining AI-driven issue diagnosis, smart worker assignment, real-time GPS tracking, ETA prediction, dynamic pricing, secure blockchain-based payments, and continuous monitoring into a single integrated platform. By reducing response time, improving decision-making, enhancing transparency, and optimizing resource allocation, INVOLVE delivers a reliable, efficient, and user-centric solution that addresses the shortcomings of traditional local service platforms while meeting the growing demand for rapid emergency assistance.

5. Result Analysis

The proposed **INVOLVE – AI-Powered On-Demand Local Services Platform** was successfully designed, developed, and evaluated to determine its effectiveness in providing intelligent and efficient emergency local service management. The platform integrates Artificial Intelligence (AI), real-time communication, GPS-based tracking, intelligent worker recommendation, ETA prediction, dynamic pricing, and blockchain-secured digital payments into a unified system. The primary objective of the evaluation was to verify whether the proposed platform could overcome the limitations of traditional service booking applications by reducing response time, improving worker assignment accuracy, enhancing service reliability, and providing greater transparency throughout the service lifecycle.

Functional testing was performed on all major modules of the platform under different emergency service scenarios, including electrical faults, plumbing issues, vehicle breakdowns, appliance repair, and home maintenance requests. The testing confirmed that the system successfully executes the complete service workflow, beginning with user registration and request submission, followed by AI-based issue diagnosis, intelligent worker assignment, real-time communication, service tracking, secure payment processing, and customer feedback collection. Users were able to submit service requests through text descriptions and image uploads, enabling the AI engine to analyse the reported issue and determine the appropriate service category and emergency priority.

The AI-based diagnosis module demonstrated effective performance by reducing communication gaps between customers and professionals. By analysing textual descriptions and uploaded images, the system identified the probable issue before assigning a worker. This enabled professionals to understand the service requirements in advance and prepare the necessary tools and equipment before reaching the customer's location. Consequently, unnecessary repeat visits were minimized, improving first-time service completion and overall service efficiency.

The intelligent worker recommendation module produced better assignment results than traditional location-based matching systems. Instead of assigning the nearest available professional, the platform evaluated multiple decision parameters, including worker expertise, certification, customer ratings, historical performance, current workload, availability, travel distance, and emergency priority. Based on these factors, the AI generated a suitability score for each professional and selected the most appropriate worker. This approach improved assignment accuracy, balanced workload distribution, and increased the likelihood of successful service completion within the shortest possible time.

The real-time communication and GPS tracking modules also demonstrated reliable performance during evaluation. Customers received continuous notifications regarding request status, worker assignment, estimated arrival time, and service completion. Live GPS tracking enabled users to monitor the movement of the assigned professional throughout the service journey. The ETA prediction module combined GPS data with traffic conditions to provide accurate arrival estimates, reducing customer uncertainty and improving transparency during emergency situations.

The dynamic pricing module successfully calculated service charges based on multiple parameters, including service category, issue complexity, travel distance, emergency priority, and estimated repair effort. Transparent pricing was displayed before booking confirmation, allowing customers to understand the expected cost in advance and reducing billing disputes. After successful service completion, the blockchain-enabled payment module securely processed digital transactions while maintaining tamper-resistant payment records. This enhanced financial security and increased trust between customers and service professionals.

Overall, the proposed platform demonstrated significant improvements compared with conventional local service applications. The integration of Artificial Intelligence enabled intelligent issue diagnosis and worker recommendation, while real-time communication improved coordination among users, professionals, and administrators. GPS tracking and ETA prediction enhanced transparency, and blockchain-based payment verification ensured secure financial transactions. The platform achieved faster emergency response, improved worker utilization, higher assignment accuracy, better service reliability, and increased customer satisfaction. These results validate that the proposed methodology provides an efficient, scalable, and user-centric solution for modern on-demand local emergency services.

6. DISCUSSION

The implementation and evaluation of INVOLVE – An AI-Powered On-Demand Local Services Platform demonstrate the effectiveness of integrating Artificial Intelligence with modern communication and location-based technologies to improve emergency local service delivery. The results indicate that AI-based issue diagnosis and intelligent worker recommendation significantly enhance the efficiency of service allocation compared with conventional platforms that primarily rely on manual booking or proximity-based assignment. By considering factors such as worker expertise, availability, customer ratings, workload, emergency priority, and travel distance, the system is able to assign the most suitable professional for each request.

The integration of real-time GPS tracking and ETA prediction improves transparency by allowing users to monitor service progress and receive accurate arrival estimates. Dynamic pricing further enhances customer trust by providing clear and fair cost estimation before service confirmation, while blockchain-based payment verification ensures secure and tamper-resistant digital transactions. These features collectively contribute to improved customer satisfaction and operational efficiency.

Although the proposed platform demonstrates promising performance, certain challenges remain. The accuracy of AI-based issue diagnosis depends on the quality of user-provided descriptions and uploaded images. In addition, reliable internet connectivity and GPS availability are essential for real-time communication and location tracking. Future improvements may include multilingual AI support, voice-based diagnosis, predictive demand forecasting, IoT-based fault detection, and advanced machine learning models for more accurate worker recommendation. Overall, the proposed system provides a scalable, intelligent, and reliable solution that addresses the shortcomings of existing local service platforms while meeting the growing demand for rapid emergency assistance.

7. CONCLUSION AND FUTURE WORK

The proposed **INVOLVE – An AI-Powered On-Demand Local Services Platform** successfully demonstrates the potential of integrating Artificial Intelligence, real-time communication, geolocation technologies, and secure digital payments to improve emergency local service management. The platform addresses the major limitations of conventional service booking systems by providing AI-based issue diagnosis, intelligent worker recommendation, emergency-priority handling, real-time GPS tracking, ETA prediction, dynamic pricing, and blockchain-enabled payment verification. Instead of assigning the nearest available worker, the system evaluates multiple factors, including expertise, availability, workload, customer ratings, and emergency severity, to assign the most suitable verified professional. This intelligent approach reduces response time, improves worker utilization, enhances service reliability, and increases customer satisfaction. The modular architecture also ensures scalability, flexibility, and easy integration of future technologies, making the platform suitable for deployment across different cities and service categories.

Future work will focus on enhancing the intelligence and capabilities of the platform through advanced machine learning algorithms that continuously learn from historical service data to improve worker recommendation and demand forecasting. Predictive analytics can be incorporated to identify high-demand regions and optimize workforce planning. Additional enhancements include multilingual voice-based AI assistants, IoT integration for automatic fault detection and service request generation, AI-powered fraud detection for identifying suspicious activities, and dedicated Android and iOS applications. Future versions may also support offline emergency assistance, wearable device integration, and smart navigation using live traffic analysis. These improvements will enable **INVOLVE** to evolve into a comprehensive, intelligent, scalable, and user-centric emergency service ecosystem capable of delivering faster, safer, and more efficient local services.

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